BIO 102 LECTURE NOTE FROM DR. OKOLIE

FLOWERING AND NON-FLOWERING PLANTS

FLOWERING PLANTS

Flowering plants make up the largest and most diverse group of plants on Earth. They have colonized the majority of the planet, including the oceans and sea floors. As a result, these flowering plants, which are referred to as "angiosperms," are very important to human life. Not only are they responsible for producing a significant portion of the oxygen that we breathe, but they also produce fruit, vegetables, herbs, and seeds. They contribute to the ecology by producing things, whereas we do so by consuming things.

Flowering plants are called Angiosperms because their seeds are enclosed within a fruit. The term “angio” means vessel or enclosure, and “Sperm” means seed. Flowering plants produce flowers and seeds enclosed in the fruit. The flower is the reproductive structure containing the pollen, ovule, stamen and carpel in angiosperms. After fertilization, the ovary of the flower develops into a fruit that surrounds the seed. Flowering plants have well-developed vascular system (xylem and phloem) for nutrient transport.

NON-FLOWERING PLANTS

Non-flowering plants are those plants that do not produce flowers because they lack the specialized reproductive apparatus, which is the flower. The evolution of these plants occurred a very long time before anthophytes. In general, these plants are a lot simpler to care for than flowering plants.

Non-flowering plants reproduce by means of spores or naked seeds in cones. They include the bryophytes (Mosses), pteridophytes (ferns), and the gymnosperms (pines). Bryophytes and pteridophytes reproduce by spores, while gymnosperms reproduce by means of naked or exposed seeds in cones. Gymnosperms are a more primitive lineage than angiosperms, and evidence suggests angiosperms actually evolved from gymnosperms. The rest of the nonflowering plants are more primitive than gymnosperms, and they do not produce seeds. These are known as seedless vascular plants and seedless nonvascular plants. The word gymnosperm literally means 'naked seed,' and this refers to the fact that the seeds lack a protective coating. Instead, the cone can act as a protective structure for the seeds, sometimes preserving them for many years before releasing them.

Many gymnosperms have thick, woody stems (Pine trees and cycads (cycad palms)). Leaves are often needle-like or scale-like in shape or morphology.

Difference Between Flowering and Non-Flowering Plants

(i) Flowering plants possess flower as the reproductive structure; nonflowering plants do not possess flower.
(ii) Flowering plants produce seeds which are enclosed in fruits; non-flowering plants (gymnosperms) produce naked seeds which are carried in cones.
(iii) Flowering plants have xylem packed with vessels, but the xylem of non-flowering plants is devoid of such structures.
(iv) Flowering plants have phloem with sieve tubes and companion cells; non-flowering plants do not.

(v) Flowering plants are the only plants in which the ovule grows within the ovary; non-flowering plants do not have this characteristic.

(vi) Flowering plants do not require either external water or internal fluids to be fertilized. In contrast, non-flowering plants, particularly more primitive non-flowering plants, require external water for fertilization, whereas more developed non-flowering plants require at least internal fluids for fertilization. As a result, the spermatozoids found in flowering plants are immobile, whereas the spermatozoids found in non-flowering plants are mostly mobile.

CIRCULATORY SYSTEM IN PLANTS

The circulatory system in plants is also known as the vascular system in plants. The vascular system allows plants to grow taller and larger, enabling them to inhabit a wide variety of environments. Without these conduits, plants only grow to a small size. Non-vascular plants, such as mosses and liverworts, lack xylem and phloem and rely on diffusion and osmosis for the distribution of nutrients. Vascular plants, including trees, flowering plants, and ferns, use xylem and phloem to efficiently transport nutrients, even against gravity.

XYLEM

The term "xylem" comes from the Greek word "xylon," which means "wood." This reflects the role of xylem tissue in contributing to the structural strength of plants, particularly woody ones.

Function and Structure of Xylem

Xylem transports water and dissolved minerals absorbed from the soil by the roots to the above-ground parts of the plant. The plant uses the water transported by the xylem in both photosynthesis and transpiration. Additionally, the xylem also provides structural support to the plant.

The xylem tissue consists of four main types of cells: tracheids, vessel elements, xylem parenchyma, and xylem fibers. The vessel elements and tracheids are the water-conducting cells. Vessel elements are wider and shorter than tracheids and connect together at the ends. The ends have perforation plates that permit water transfer between cells. Tracheids are long, thin, and tapered at the ends. The secondary cell walls of the tracheids contain lignin. The parenchyma stores food and helps in the repair and growth of xylem, while xylem fibers provide support.

In most plants, the xylem is in the center of the stem, forming a core of rigid, woody material. Mature xylem consists of dead vessel element and tracheid cells connected by hollow ends.

Transportation in Xylem

The mechanism of water transport in xylem primarily involves a process known as cohesion-tension theory. Here, the evaporative pull of transpiration from the leaves creates a tension or negative pressure that pulls water upward from the roots through the xylem tissue. Also, root pressure also plays a role. Here, water enters roots from the soil via osmosis, generating a positive pressure that forces water upward into the plant. Capillary action, adhesion, cohesion, and differences in water potential also contribute to water movement through xylem

PHLOEM

The term “phloem” comes from the Greek word “phloios,” meaning “bark.” This name is fitting, as phloem is often found just beneath the bark in trees.

Function and Structure of Phloem

Phloem transports organic nutrients, particularly sugars synthesized during photosynthesis, from the leaves to all other cells of the plant, including the roots.

Phloem tissue is composed of sieve-tube elements, companion cells, phloem fibers, and phloem parenchyma. The sieve-tube elements, along with their companion cells, primarily control the transportation of food. Phloem fibers provide support, and phloem parenchyma assists with food storage and the secretion of plant resins.

In most plants, the phloem is towards the exterior of the plant, just below the bark in stems and roots. The sieve-tube cells are alive, but they lack a nucleus and have less cytoplasm than other plant cells. The companion cells are living cells with a normal composition.

Transportation in Phloem

The transport mechanism in phloem is known as translocation. It involves an active process where sugars load into sieve tubes in the leaves (source) and unload where they are needed (sink), such as roots or developing shoots. This differential in sugar concentration results in water moving from xylem to phloem, building a pressure that drives the sap down the plant.

Xylem and Phloem Working Together

Although xylem and phloem serve different functions, they operate in close coordination to maintain the health and growth of the plant. Together, they form the vascular system that distributes vital substances throughout the organism, much like the circulatory system in animals.

The xylem transports water and mineral nutrients from the roots upward to the leaves, supporting photosynthesis and replacing water lost through transpiration. These processes generate the sugars and other organic compounds that the phloem then distributes from the leaves to other plant parts, including growing stems, roots, fruits, and flowers.

This cooperative relationship is evident in several ways:

(i) Water-Sugar Interdependence: The water carried by xylem is essential for photosynthesis in the leaves, which in turn produces the sugars transported by phloem.

(ii) Pressure Gradient Interaction: The high concentration of sugars in phloem sieve tubes causes water to enter from the xylem by osmosis, increasing turgor pressure and facilitating the downward flow of sap in phloem (translocation).

(iii) Resource Allocation: The phloem supplies energy-rich compounds to roots, which rely on that nourishment to grow and continue absorbing water and minerals for xylem transport.

Ultimately, the synergy between xylem and phloem helps vascular plants grow tall, store nutrients, adapt to environmental challenges, and reproduce successfully. The two systems are interdependent and indispensable for plant life.

Differences in Xylem and Phloem in Monocots and Dicots

Monocots and dicots differ in the arrangement and structure of their xylem and phloem.

In dicot plants, the vascular system is organized in a ring, with the xylem typically inside, surrounded by phloem. There is often a region of meristematic cambium cells, which divide to produce more xylem or phloem cells, allowing the stem or root to increase in diameter.

In monocot plants, the xylem and phloem are paired into bundles scattered throughout the stem. Monocots do not have a vascular cambium, meaning they typically do not increase in diameter after growth.

Girdling and Its Impacts

Girdling is a practice that removes a ring of bark (the phloem layer) from around the entire circumference of a tree or plant stem. This disrupts the downward transportation of sugars and other metabolites from the leaves through the phloem. Girdling can cause the death of a tree because it interrupts the supply of food from leaves to the roots, essentially starving the plant.

However, girdling also has a deliberate use in horticulture. It encourages the plant to produce larger fruits or to direct the plant's energy towards certain branches. By disrupting the flow of nutrients, the plant overcompensates in the remaining portions, often leading to increased yield or size of the produce.

XYLEM AND PHLOEM TERMS

Bast fibers
Supportive fibers found in the phloem that add mechanical strength to the plant.

Cambium
A layer of meristematic cells between xylem and phloem in dicots that divides to produce new vascular tissue and allows secondary growth (increase in diameter).

Capillary action
The ability of water to flow in narrow spaces without external forces, aiding water movement in xylem.

Cohesion-tension theory
The explanation for water movement in xylem: transpiration at the leaves pulls water upward due to water's cohesive properties and the tension generated.

Companion cells
Living cells adjacent to sieve tube elements in phloem that regulate sugar transport and perform metabolic functions for the sieve tubes.

Girdling
The practice of removing a ring of bark (phloem) from a plant or tree trunk, disrupting phloem transport and potentially killing the plant.

Lignin
A complex organic polymer in cell walls that provides rigidity and waterproofing, especially in xylem vessels and fibers.

Meristematic tissue
Undifferentiated plant tissue capable of cell division; the vascular cambium is an example.

Phloem
The living vascular tissue that transports organic nutrients, such as sugars, from the leaves (source) to other parts of the plant (sink).

Phloem parenchyma
Living cells in the phloem involved in food storage and secretion.

Phloem sap
The fluid transported in phloem, rich in sugars, hormones, amino acids, and other organic molecules.

Sieve plate
A porous structure at the ends of sieve tube elements that allows the flow of phloem sap between cells.

Sieve tube elements
Elongated living cells in phloem that transport sugars but lack nuclei and have limited cytoplasm; they rely on companion cells.

Turgor pressure
The internal water pressure within a cell that provides structural support and drives movement in phloem transport.

Translocation
The active movement of organic substances (mainly sugars) through phloem from source to sink.

Tracheids
Long, tapered cells in xylem that conduct water and provide support; water passes through pits in the cell walls.

Vascular bundle
A strand of xylem and phloem tissues (often with cambium) running through the stems, roots, and leaves of vascular plants.

Vessel elements
Short, wide xylem cells with perforation plates that align end-to-end to form continuous tubes for efficient water conduction.

Xylem
The non-living vascular tissue that conducts water and dissolved minerals from roots to aerial parts of the plant and provides structural support.

Xylem parenchyma
Living xylem cells that store nutrients and assist in repair and lateral transport.

Xylem sap
The dilute fluid (mostly water and minerals) transported upward through xylem vessels and tracheids.

CIRCULATORY SYSTEM IN ANIMALS

In all animals, except a few simple types, the circulatory system is used to transport nutrients and gases through the body. Simple diffusion allows some water, nutrient, waste, and gas exchange into primitive animals that are only a few cell layers thick; however, bulk flow is the only method by which the entire body of larger more complex organisms is accessed.

Circulatory System Architecture

The circulatory system is effectively a network of cylindrical vessels: the arteries, veins, and capillaries that emanate from a pump, the heart. In all vertebrate organisms, as well as some invertebrates, this is a closed-loop system, in which the blood is not free in a cavity. In a closed circulatory system, blood is contained inside blood vessels and circulates unidirectionally from the heart around the systemic circulatory route, then returns to the heart again. As opposed to a closed system, arthropods including insects, crustaceans, and most mollusks, have an open circulatory system. In an open circulatory system, the blood is not enclosed in the blood vessels but is pumped into a cavity called a hemocoel and is called hemolymph because the blood mixes with the interstitial fluid. As the heart beats and the animal moves, the hemolymph circulates around the organs within the body cavity and then reenters the hearts through openings called ostia. This movement allows for gas and nutrient exchange. An open circulatory system does not use as much energy as a closed system to operate or to maintain; however, there is a trade-off with the amount of blood that can be moved to metabolically active organs and tissues that require high levels of oxygen. In fact, one reason that insects with wing spans of up to two feet wide (70 cm) are not around today is probably because they were outcompeted by the arrival of birds 150 million years ago. Birds, having a closed circulatory system, are thought to have moved more agilely, allowing them to get food faster and possibly to prey on the insects.

The earthworm possesses a closed circulatory system of an earthworm. In an earthworm, dorsal and ventral blood vessels run along the top and bottom of the intestine, respectively. The dorsal and ventral blood vessels are connected by ring-like hearts. Hearts are also associated with the dorsal blood vessel. These hearts pump blood forward, and the ring-like hearts pump blood down to the ventral vessel, which returns blood to the back of the body. The bee has an open circulatory system. The dorsal blood vessel, which contains multiple hearts, runs along the top of the bee. Blood exits the dorsal blood vessel through an opening in the head, into the body cavity. Blood reenters the blood vessels through openings in the hearts called ostia.

In (a) closed circulatory systems, the heart pumps blood through vessels that are separate from the interstitial fluid of the body. Most vertebrates and some invertebrates, like this annelid earthworm, have a closed circulatory system. In (b) open circulatory systems, a fluid called hemolymph is pumped through a blood vessel that empties into the body cavity. Hemolymph returns to the blood vessel through openings called ostia. Arthropods like this bee and most mollusks have open circulatory systems.

Circulatory System Variation in Animals

The circulatory system varies from simple systems in invertebrates to more complex systems in vertebrates. The simplest animals, such as the sponges (Porifera) and rotifers (Rotifera), do not need a circulatory system because diffusion allows adequate exchange of water, nutrients, and waste, as well as dissolved gases. Organisms that are more complex but still only have two

layers of cells in their body plan, such as jellies (Cnidaria) and comb jellies (Ctenophora) also use diffusion through their epidermis and internally through the gastrovascular compartment. Both their internal and external tissues are bathed in an aqueous environment and exchange fluids by diffusion on both sides. Exchange of fluids is assisted by the pulsing of the jellyfish body.

For more complex organisms, diffusion is not efficient for cycling gases, nutrients, and waste effectively through the body; therefore, more complex circulatory systems evolved. Most arthropods and many mollusks have open circulatory systems. In an open system, an elongated beating heart pushes the hemolymph through the body and muscle contractions help to move fluids. The larger more complex crustaceans, including lobsters, have developed arterial-like vessels to push blood through their bodies, and the most active mollusks, such as squids, have evolved a closed circulatory system and are able to move rapidly to catch prey. Closed circulatory systems are a characteristic of vertebrates; however, there are significant differences in the structure of the heart and the circulation of blood between the different vertebrate groups due to adaptation during evolution and associated differences in anatomy.

Figure 1. The basic circulatory systems of some vertebrates: fish, amphibians, reptiles, and mammals.

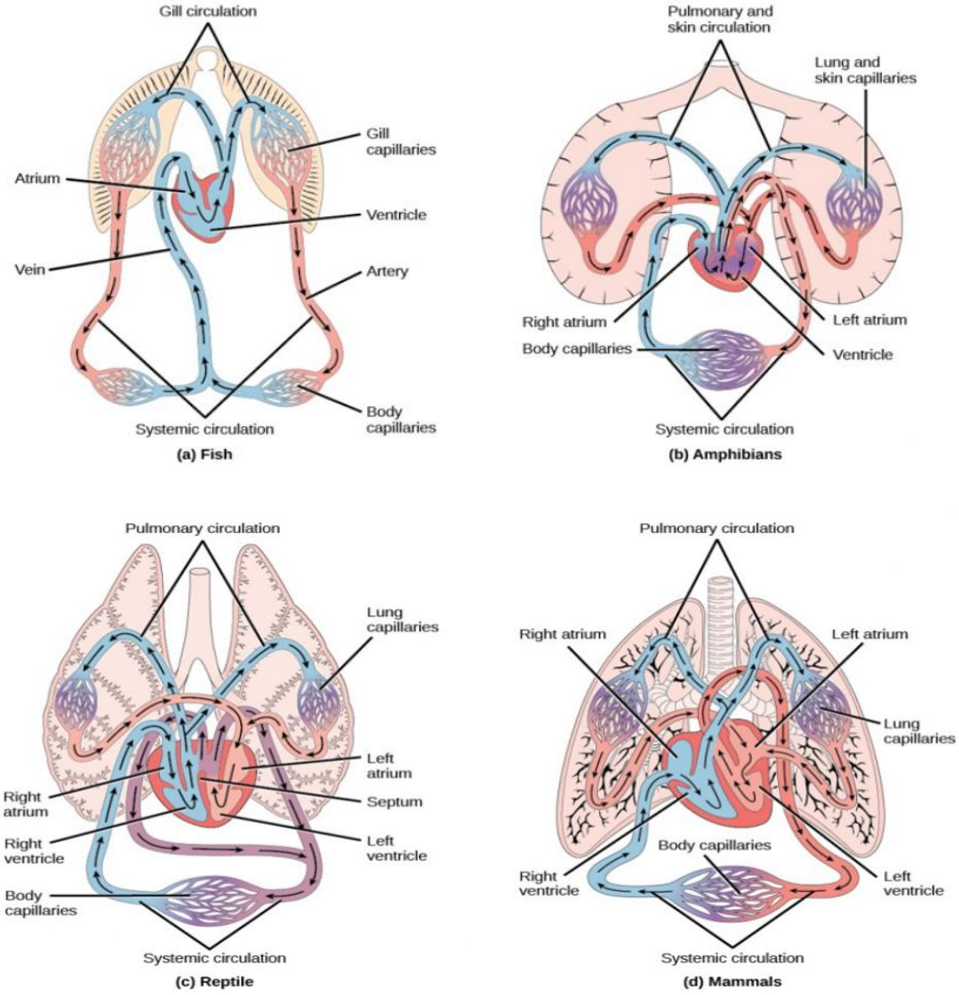


Illustration (a) shows the circulatory system of fish, which have a two-chambered heart with one atrium and one ventricle. Blood in systemic circulation flows from the body into the atrium,

then into the ventricle. Blood exiting the heart enters gill circulation, where gases are exchanged by gill capillaries. From the gills blood re-enters systemic circulation, where gases in the body are exchanged by body capillaries Fish have the simplest circulatory systems of the vertebrates: blood flows unidirectionally from the two-chambered heart through the gills and then the rest of the body. Illustration (b) shows the circulatory system of amphibians, which have a three-chambered heart with two atriums and one ventricle. Blood in systemic circulation enters the heart, flows into the right atrium, then into the ventricle. Blood leaving the ventricle enters pulmonary and skin circulation. Capillaries in the lung and skin exchange gases, oxygenating the blood. From the lungs and skin blood re-enters the heart through the left atrium. Blood flows into the ventricle, where it mixes with blood from systemic circulation. Blood leaves the ventricle and enters systemic circulation. Illustration (c) shows the circulatory system of reptiles, which have a four-chambered heart. The right and left ventricle are separated by a septum, but there is no septum separating the right and left atrium, so there is some mixing of blood between these two chambers. Blood from systemic circulation enters the right atrium, then flows from the right ventricle and enters pulmonary circulation, where blood is oxygenated in the lungs. From the lungs blood travels back into the heart through the left atrium. Because the left and right atrium are not separated, some mixing of oxygenated and deoxygenated blood occurs. Blood is pumped into the left ventricle, then into the body. Illustration (d) shows the circulatory system of mammals, which have a four-chambered heart. Circulation is similar to that of reptiles, but the four chambers are completely separate from one another, which improves efficiency.

Amphibians have two circulatory routes: one for oxygenation of the blood through the lungs and skin, and the other to take oxygen to the rest of the body. The blood is pumped from a three-chambered heart with two atria and a single ventricle. Figure (c): Reptiles also have two circulatory routes; however, blood is only oxygenated through the lungs. The heart is three chambered, but the ventricles are partially separated so some mixing of oxygenated and deoxygenated blood occurs except in crocodilians and birds. Figure (d): Mammals and birds have the most efficient heart with four chambers that completely separate the oxygenated and deoxygenated blood; it pumps only oxygenated blood through the body and deoxygenated blood to the lungs.

As illustrated in Figure 1(a), Fish have a single circuit for blood flow and a two-chambered heart that has only a single atrium and a single ventricle. The atrium collects blood that has returned from the body and the ventricle pumps the blood to the gills where gas exchange occurs and the blood is re-oxygenated; this is called gill circulation. The blood then continues through the rest of the body before arriving back at the atrium; this is called systemic circulation. This unidirectional flow of blood produces a gradient of oxygenated to deoxygenated blood around the fish's systemic circuit. The result is a limit in the amount of oxygen that can reach some of the organs and tissues of the body, reducing the overall metabolic capacity of fish.

In amphibians, reptiles, birds, and mammals, blood flow is directed in two circuits: one through the lungs and back to the heart, which is called pulmonary circulation, and the other throughout the rest of the body and its organs including the brain (systemic circulation). In amphibians, gas exchange also occurs through the skin during pulmonary circulation and is referred to as pulmocutaneous circulation. As shown in Figure (b), amphibians have a three-chambered heart that has two atria and one ventricle rather than the two-chambered heart of fish. The two atria

(superior heart chambers) receive blood from the two different circuits (the lungs and the systems), and then there is some mixing of the blood in the heart's ventricle (inferior heart chamber), which reduces the efficiency of oxygenation. The advantage to this arrangement is that high pressure in the vessels pushes blood to the lungs and body. The mixing is mitigated by a ridge within the ventricle that diverts oxygen-rich blood through the systemic circulatory system and deoxygenated blood to the pulmocutaneous circuit. For this reason, amphibians are often described as having double circulation.

Most reptiles also have a three-chambered heart similar to the amphibian heart that directs blood to the pulmonary and systemic circuits, as shown in Figure (c). The ventricle is divided more effectively by a partial septum, which results in less mixing of oxygenated and deoxygenated blood. Some reptiles (alligators and crocodiles) are the most primitive animals to exhibit a four-chambered heart. Crocodilians have a unique circulatory mechanism where the heart shunts blood from the lungs toward the stomach and other organs during long periods of submergence, for instance, while the animal waits for prey or stays underwater waiting for prey to rot. One adaptation includes two main arteries that leave the same part of the heart: one takes blood to the lungs and the other provides an alternate route to the stomach and other parts of the body. Two other adaptations include a hole in the heart between the two ventricles, called the foramen of Panizza, which allows blood to move from one side of the heart to the other, and specialized connective tissue that slows the blood flow to the lungs. Together these adaptations have made crocodiles and alligators one of the most evolutionarily successful animal groups on earth.

In mammals and birds, the heart is also divided into four chambers: two atria and two ventricles, as illustrated in Figure (d). The oxygenated blood is separated from the deoxygenated blood, which improves the efficiency of double circulation and is probably required for the warm-blooded lifestyle of mammals and birds. The four-chambered heart of birds and mammals evolved independently from a three-chambered heart. The independent evolution of the same or a similar biological trait is referred to as convergent evolution.